

Mestrado em Engenharia Eletrotécnica e de Computadores

Gestão de Projetos de Engenharia

Mobile Communications' Base Stations and Wi-Fi Access Points inside a Cruise Ship

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Abstract

This project developed a planning package for an onboard telecommunications system for the MV World Voyager. It covered the technical design, schedule, risk, quality, and cost planning needed to support reliable cellular and Wi-Fi service at sea and in port.

The study defined the coverage and capacity plan, selected marine-grade equipment, and outlined the backhaul and telecom-room architecture. The result is a realistic and compliant plan that can support future implementation on board the vessel.

1 Introduction and Project Context

1.1 Project rationale

Modern cruise passengers expect continuous mobile and Wi-Fi connectivity, even at sea, and operators use connectivity to differentiate services and support onboard operations. At the same time, maritime communications must cope with harsh propagation conditions inside metal ships, rely on satellite backhaul offshore, and avoid interference with terrestrial mobile networks in port.

This project addresses these challenges by designing and planning an onboard telecommunications system for the *MV World Voyager*. The goal is to define a technically feasible and compliant solution that improves connectivity for passengers and crew while respecting the ship's operational and aesthetic constraints.

1.2 Selected cruise ship

The *MV World Voyager* is a small luxury cruise ship with around 200 passengers and 120 crew members distributed across six main decks [1]. Its metallic structure, compartmentalised layout and strict interior design standards make wireless coverage planning non-trivial [2]. The ship spends significant time at sea, where satellite connectivity is essential, and regularly visits ports where emissions must be controlled [2].

For this project, it is assumed that deck layouts and basic technical documentation are available, including room locations, existing telecommunications rooms and feasible cable routes. This information supports later decisions on base station placement, Wi-Fi access point locations and cabling paths.

1.3 Project objectives

The primary objective is to design and plan an onboard telecommunications system for the *MV World Voyager* that provides reliable cellular and Wi-Fi coverage for passengers and crew, using satellite backhaul at sea and a shutdown mechanism in port. More specifically, the project aims to:

- Define a coverage and capacity plan for cellular and Wi-Fi services across cabins, public areas, exterior decks and key operational spaces.
- Select suitable indoor base stations, antennas, Wi-Fi access points and network equipment for a maritime environment, including satellite backhaul integration and port shutdown behaviour.

From a SMART perspective, the solution must cover the six passenger decks and main operational zones, support the expected simultaneous usage of roughly 325 people, rely on realistic maritime-grade equipment and services, add value to the ship's connectivity offer while complying with regulations, and be delivered by mid-June 2026.

1.4 Scope, assumptions and constraints

The scope of the project is limited to the planning and conceptual design of the onboard telecommunications system. It includes radio-frequency planning, selection and positioning of base stations, antennas and Wi-Fi access points, high-level design of cabling and equipment rooms, and the definition of satellite backhaul integration and port shutdown logic.

The work relies on a few key assumptions: the ship owner provides timely access to deck plans and technical documentation, suppliers and the satellite provider can offer typical marine-grade solutions compatible with the designed topology, and regulatory inputs are available at a level sufficient to define a shutdown mechanism and basic compliance checks.

The main constraints are the use of only three labor cost categories (80, 100 and 120 €/hour) for budgeting, the need for a cellular shutdown capability in port to avoid interference with terrestrial networks, and the physical and aesthetic limitations imposed by the ship's metal structure and luxury interior design.

1.5 Completion criteria

The project is considered complete when the proposed telecommunications solution has been successfully validated against the agreed operational requirements and is ready for deployment on board the *MV World Voyager*. Completion therefore depends on the solution meeting the required coverage, capacity, backhaul, shutdown, and compliance conditions, and on the main technical risks having been addressed through the design.

In practical terms, this means that the system design must be technically feasible, compliant with the applicable maritime and regulatory constraints, and supported by a clear validation approach. The project is only complete once the proposed solution can be confidently taken forward to implementation and acceptance on the vessel.

2 Stakeholders

The success of the World Voyager connectivity project depends on the strategic alignment and active participation of a diverse group of internal and external Stakeholders. Their identification and categorization (Table 1) is vital so as to pinpoint possible liabilities and roadblocks.

Table 1: Project Stakeholder Analysis & Management Plan

Stakeholder	Type	Att.	Interests	Potential impact	Expectations	Attitudes and/or risk
Project Leader & Project Development Dept. (S1)	Int	+	Successful delivery of the network infrastructure within deadlines and allocated budget.	Critical: Controls the WBS, resource allocation, and daily management of the critical path.	Project Management Plan, coordination between all internal/external parties, and technical reporting.	Highly Committed: Goal-oriented. Risk: Potential over-allocation of resources or delays in dependency tasks.
Atlas Ocean Voyages (Customer / Ship Owner) (S2)	Ext	±	Provide premium connectivity to passengers to stay competitive; ensure ship safety and aesthetic integrity.	High: Can grant or deny access to the vessel; must approve all physical modifications (drilling, cabling).	Access to deck plans, technical specifications of the ship, and installation "windows of opportunity."	Supportive but Protective: Cautious about disruptions to cruise schedules or damage to ship interiors.
Mystic Invest Holding (COO / Sponsor) (S3)	Ext	±	Ensure the project adds value to the fleet and stays within the corporate financial framework.	High: Has ultimate veto power; can cancel the project, change the budget, or re-prioritize resources.	Financial backing, final approval of the project charter, and high-level strategic direction.	Supportive but Cost-Conscious. Risk: Might cut funding if labor or equipment costs escalate.
Radio Engineering Dept. (Internal Tech) (S4)	Int	+	Ensure network architecture is sound, frequencies optimized, and 100% coverage across all passenger decks.	High: Technical studies determine the number of base stations. Errors lead to "dead zones" or dropped calls.	Coverage heat-maps, frequency planning, equipment specs, and final technical validation.	Professional and Technical. Risk: May request expensive equipment that conflicts with budget.
Regulation Dept. (Internal Compliance) (S5)	Int	N	Ensuring the system follows ITU and National Telecommunications Authorities guidelines.	High: If they don't secure the right to use specific frequencies, the system cannot legally broadcast.	Licensing for satellite backhaul, frequency permits, and technical compliance reports.	Compliance-Focused. Risk: Fines or ship detention if frequencies interfere with local networks.
Management Dept. (Finance/Admin) (S6)	Int	±	Strict control over project expenditures and optimization of human resource allocation.	High: Can freeze the budget or demand a reduction in team size if ROI decreases.	Approval of the financial plan, authorization for high-cost labor, and budget reconciliation.	Economical and Result-Oriented. Risk: May reject technical expenses exceeding estimates.
Law Dept. (Internal Legal) (S7)	Int	N	Protecting the operator from legal liability and ensuring roaming contracts are enforceable.	High: Must sign off on all contracts with Atlas Ocean Voyages and satellite providers.	Draft and review of Service Level Agreements (SLAs), roaming contracts, and liability waivers.	Risk-Averse: Thorough but slow. Risk: Legal bottlenecks if contracts misalign with maritime law.
Buying Dept. (Internal Procurement) (S8)	Int	+	Staying within the project budget while ensuring high-quality hardware is acquired.	Medium/High: Ability to negotiate prices affects the "Financial Management" evaluation of the project.	Purchase orders for hardware; management of delivery timelines and vendor negotiations.	Cost-Conscious. Risk: Long lead times for specialized marine-grade hardware.
Marketing Dept. (Internal Commercial) (S9)	Int	+	Create attractive service packages to maximize revenue and passenger satisfaction.	Medium: Defines the User Experience requirements (e.g. 4K streaming), which dictates technical design limits.	Service definitions, pricing models, branding for the "Ship Portal," and promotional material.	Enthusiastic. Risk: May promise features/speeds that the satellite backhaul cannot support.
Ship Captain (Terje Ulset) (S10)	Ext	N	Safety of the vessel and passengers; ensuring no interference with bridge navigation/radar.	High (Operational): Can deny installation work or power-on tests if he perceives a safety risk.	Final on-site authorization for work, access to technical spaces, and coordination of "Quiet Hours."	Pragmatic and Safety-First. Risk: May halt work during bad weather or critical maneuvers.
Constructor company (S11)	Ext	+	Complete the physical installation according to contract to maintain reputation with Atlas.	High (Execution): Speed and quality determine the physical "Go-Live". Poor work leads to system failure.	Skilled labor, specialized tools for maritime environments, and "As-Built" documentation.	Task-Oriented. Risk: May take shortcuts in hidden areas or experience access delays.
Satellite Link Supplier (S12)	Ext	+	Provide bandwidth services and maintain long-term contract for maritime data traffic.	Critical (Technical): If the link fails or lacks coverage in polar regions, the project value drops to zero.	High-speed backhaul, low-latency connection, and technical support for tracking antennas.	Service-Oriented. Risk: Signal degradation in extreme weather or polar blind spots.
Equipment Suppliers (S13)	Ext	+	To sell high-end telecommunications hardware and secure maintenance contracts.	Medium: Faulty hardware or lack of maritime certs causes delays in the commissioning phase.	Marine-grade Wi-Fi 6 access points, cellular small cells, cabling, and technical manuals.	Commercial and Supportive. Risk: Hardware incompatibility with ship's power grid.
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Stakeholder (name and role)	Type	Att.	Interest	Potential impact	Expectations	Attitudes and/or risk
Cybersecurity Consultancy (S14)	Ext	±	Secure the passenger network against cyber threats and ensure GDPR/data-protection compliance for onboard traffic.	Medium/High: Vulnerabilities in the Wi-Fi/cellular system could expose passenger data and damage the operator's reputation.	Penetration testing, security architecture review, incident response procedures, and GDPR compliance audits.	Positive on security but may delay rollout to enforce hardening measures. Risk: Conflicting recommendations with rapid deployment goals.
Ship Security (ISPS) (S15)	Ext	N	Maintaining ISPS (International Ship and Port Facility Security) standards and preventing unauthorized access to restricted areas.	Medium: Can restrict access to server rooms or the bridge if clearances aren't provided.	Badges for the installation team, oversight of equipment boarding, and credential verification.	Vigilant and Procedural. Risk: Delays in "Work Permits" if background checks are late.
World Voyager Crew (Rest of Crew) (S16)	Ext	+	Reliable communication with families and efficient operation of internal ship apps.	Low/Medium: Feedback during testing is vital for identifying "weak spots" in crew quarters.	Cooperation during hardware installation in service areas and reporting network performance.	Supportive and Eager. Risk: Misuse of bandwidth or interference with work duties.
World Voyager Passengers (S17)	Ext	±	High-speed Wi-Fi and 5G roaming to share photos and stay connected with family.	High (Indirect): Satisfaction scores determine commercial success and future investments.	Revenue through service subscriptions and feedback on the user interface/portal.	Demanding and High-Expectation. Risk: Negative reviews if signal drops in polar zones.
Regulatory Agencies (e.g., ANA-COM) (S18)	Ext	N	Spectrum management, ensuring no illegal frequency use, and enforcing Mobile Communications on board Vessels rules.	High (Legal): Can fine the operator or order system shutdown if interference occurs in territorial waters.	Granting of radio licenses, allocation of frequency bands, and safety certifications.	Neutral and Strict. Risk: Surprise port inspections and long lead times for approvals.
Classification Society (Det Norske Veritas) (S19)	Ext	N	Certify that the ship remains compliant with maritime structural and electrical standards after the installation.	High: Must approve any structural or electrical modification; without sign-off the ship loses its class certificate.	Technical review of drilling, cabling, and power integration plans; on-site inspection and certification update.	Neutral and Strict. Risk: Rejection of installation plans if marine-grade standards are not met.
Flag State Administration (S20)	Ext	N	Ensure onboard radio installations comply with SOLAS and the ship's flag-state radio regulations.	High (Legal): Authorizes the ship's radio licence; can suspend operations if installations are unapproved.	Approval of radio licence amendments and SOLAS-compliant radio installation certificates.	Neutral and Procedural. Risk: Slow administrative cycles delaying go-live.
Insurance Underwriter (S21)	Ext	±	Reassess the ship's risk profile after the telecommunications modification to maintain hull and liability coverage.	Medium: Can adjust premiums or refuse coverage if the modification is perceived as high-risk.	Risk assessment review, approval of modification, updated policy terms.	Cautious and Risk-Averse. Risk: Premium increases or coverage exclusions affecting project ROI.
Port Authorities (S22)	Ext	N	Safety of port operations and ensuring no interference with port communications (VHF/Radar).	Medium: Can restrict use of on-board equipment while docked if it interferes with shore networks.	Permission to maintain satellite tracking while docked and adherence to "Radio Silence" zones.	Neutral and Regulatory. Risk: Conflicting local regulations in different international ports.
Other Mobile Operators (Roaming Partners) (S23)	Ext	±	Protection of licensed spectrum and potential roaming revenue from passengers.	Medium: Can file complaints with regulators if ship signal "leaks" and interferes with their customers, or become partners via roaming.	Reciprocal roaming agreements and technical parameters for cell handovers in port.	Competitive yet potentially Collaborative. Risk: Legal disputes over spectrum "pollution" or interference.
Other Cruise Companies (Competitors) (S24)	Ext	–	Competitive benchmarking; monitoring new industry standards for passenger connectivity.	Low/Medium: Can influence regulatory bodies or pressure vendors for exclusive hardware/pricing.	Indirect benchmarking data and adherence to fair competition/non-interference in shared port frequencies.	Competitive and Watchful. Risk: May attempt to outpace service offerings to shift market share.

Legend: **Type:** Int = Internal, Ext = External | **Attitude (Att.):** + = Positive/Supportive, ± = Mixed/Neutral, N = Neutral/Procedural/Regulatory, – = Negative

3 Requirements

Since the project concerns the installation of cellular base stations and Wi-Fi access points inside the *MV World Voyager*, the requirements were grouped into business requirements, technical solution requirements, and project management requirements. This structure makes it easier to distinguish between the expected value of the project, the characteristics of the proposed solution, and the conditions under which the project can be carried out. The complete set of requirements, with their priorities and verification criteria, is summarised in Table 2.

Table 2: Business, technical, and project management requirements

ID	Type	Requirement	Priority	Verification Criteria
BR-01	Business	The solution shall improve passenger access to mobile communications and Wi-Fi services while the ship is at sea.	Must-have	Verified by the coverage scope and service objectives.
BR-02	Business	The project shall increase the attractiveness of the ship by supporting modern connectivity expectations.	Should-have	Verified by the business justification and stakeholder expectations.
BR-03	Business	The solution shall support reliable communications for passengers and relevant operational areas.	Must-have	Verified by the coverage and capacity plan.
TR-01	Technical	The system shall provide cellular and Wi-Fi coverage in cabins, public areas, exterior decks and operational spaces.	Must-have	Verified through radio-frequency planning and coverage maps.
TR-02	Technical	The cellular infrastructure shall include indoor base stations and antennas suitable for a maritime environment.	Must-have	Verified through the bill of materials and technical design.
TR-03	Technical	Wi-Fi access points shall be positioned to support high-traffic areas such as lounges, restaurants and pools.	Must-have	Verified through access point layout and capacity estimation.
TR-04	Technical	The onboard network shall be connected to the outside world through satellite backhaul.	Must-have	Verified through the satellite link design.
TR-05	Technical	The cellular system shall include a mechanism to avoid interference with terrestrial mobile operators when the ship is in port.	Must-have	Verified through regulatory analysis and shutdown procedure.
TR-06	Technical	The network equipment shall be installed in an appropriate technical or network room.	Should-have	Verified through equipment location planning.
PM-01	Project Mgmt.	The project shall be completed and delivered by mid-June 2026.	Must-have	Verified through the project schedule.
PM-02	Project Mgmt.	The project shall remain within the approved budget, including labour, equipment, and installation costs.	Must-have	Verified through the cost analysis and budget summary.
PM-03	Project Mgmt.	The project shall use only the predefined labour categories of 80 EUR/hour, 100 EUR/hour, and 120 EUR/hour.	Must-have	Verified through the effort and cost calculation.
PM-04	Project Mgmt.	The project scope shall be limited to planning, installation, integration, testing, and documentation activities.	Must-have	Verified through the scope statement and WBS.
PM-05	Project Mgmt.	The project shall depend on timely access to ship data, supplier information, and regulatory inputs.	Should-have	Verified through the assumptions and constraints register.

4 Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) decomposes the MV Voyager Telecommunications project into manageable deliverables and work packages. The activity list in Table 3 provides the letter-coded leaf tasks, and the WBS hierarchy in Figure 1 maps those codes to the project structure. This makes it easy to trace each detailed activity back to its place in the overall plan.

Table 3: Telecom project activities

Code	Activity	Code	Activity
A	Gather Ship Data and Analyse Deck Layout	N	Detailed Construction Package
B	Visit and Rank Candidate Locations	O	Access Permits, Planning and Safety Method Statements
C	Coverage and Base Station Layout	P	Logistics and Delivery Plan
D	Antenna and Cabling Paths	Q	Install Cabling, Base Stations, and Antennas
E	Wi-Fi AP Layout and Service Zone Design	R	Install Network Equipment and Wi-Fi APs
F	Backhaul and Core Interconnection	S	Service Configuration and Shutdown Integration
G	Cabling, Power and Telecom Room Design	T	System Commissioning
H	Hardware Technical Specifications	U	Radio and Wi-Fi Coverage and Load Testing
I	Prepare Bill of Materials	V	End-to-End Connectivity Test
J	Set Roaming Parameters	W	Port Shutdown Validation
K	Design Port Shutdown Logic	X	Final Technical Documents Conception
L	Regulatory Conformity Analysis	Y	Handover and Sign-off
M	External Approvals, Agreements and Compliance Matrix		

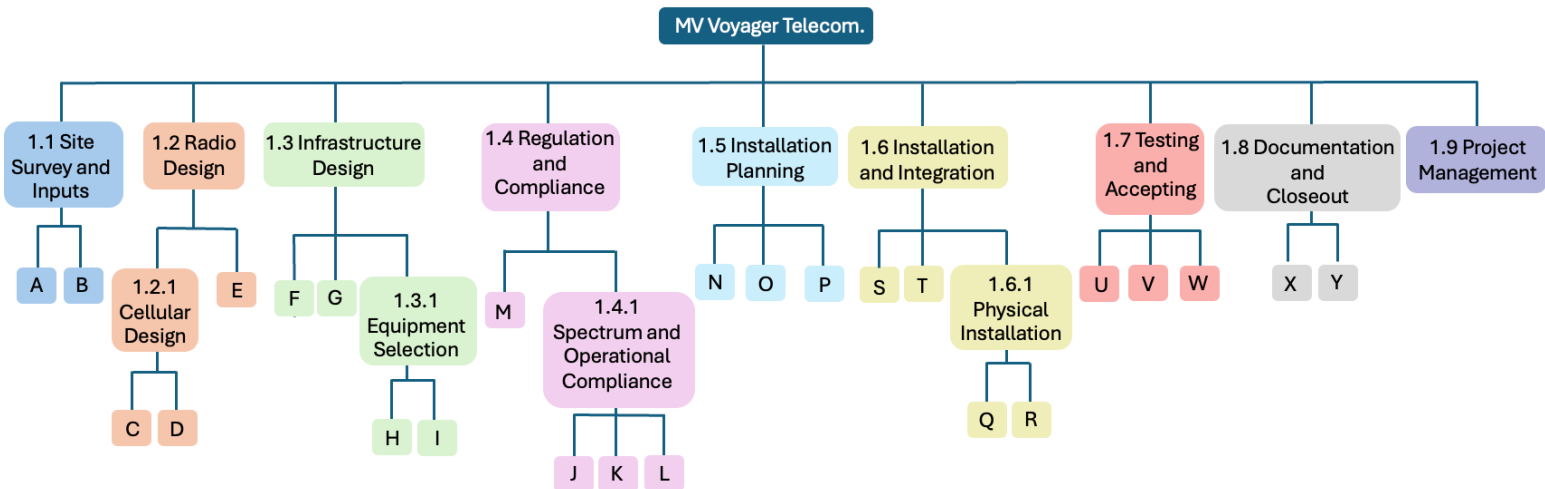


Figure 1: WBS hierarchy for MV Voyager Telecommunications Project

5 Deliverables and Milestones

5.1 Project Deliverables

The main deliverables of this project are the technical planning outputs that support the deployment of the MV World Voyager telecommunications system. They are organized to match the schedule and the work packages defined, so that each deliverable can be traced directly to a group of project activities.

- Site survey and input package, covering ship drawings, dimensions, room locations, design constraints, and candidate installation areas, corresponding to activities A and B.
- Radio design package, covering the Wi-Fi access point layout, service zones, cellular coverage concept, base-station placement, and antenna/cabling paths, corresponding to activities C, D, and E.
- Infrastructure and core network design package, covering the ship-to-shore backhaul architecture, internal interconnection, cabling routes, cable types, power feeds, and telecom room layout, corresponding to activities F and G.
- Equipment selection package, covering the marine-grade technical specifications and the bill of materials for base stations, antennas, Wi-Fi access points, cabling, accessories, and supporting components, corresponding to activities H and I.
- Regulation and compliance package, covering roaming parameters, port shutdown logic, regulatory conformity analysis, external approvals, agreements, and the compliance matrix, corresponding to activities J, K, L, and M.
- Installation planning package, covering the detailed construction package, access permits, safety method statements, and logistics and delivery plan, corresponding to activities N, O, and P.
- Installation and integration package, covering the installed cabling, mounted base stations and antennas, installed Wi-Fi APs and network equipment, service configuration, shutdown integration, and system commissioning, corresponding to activities Q, R, S, and T.
- Testing and validation package, covering radio coverage and load testing, end-to-end connectivity testing, and port shutdown validation, corresponding to activities U, V, and W.
- Final technical documentation and handover package, covering the as-built technical record, final routing and configuration information, test results, and formal handover and sign-off, corresponding to activities X and Y.

5.2 Milestones

The main project milestones, including their expected timing, responsible stakeholders and acceptance criteria, are summarize in Table 4.

Table 4: Project milestones

Milestone	Date	Responsible	Acceptance criteria
Project Kick-off	19/03/2026	Project Leader	Project formally launched and site survey/input collection activities authorized.
Design Baseline Approval	10/04/2026	Radio Engineering	Initial radio and infrastructure design baseline approved, including Wi-Fi AP layout, cellular layout, backhaul concept, and telecom room design.
Technical Package Completion	24/04/2026	Radio Engineering / Constructor	Detailed construction package completed, including work instructions, drawings, and installation notes for the following planning and implementation stages.
Regulatory Approval	23/04/2026	Regulation Department	Roaming parameters, port shutdown logic, regulatory conformity analysis, and compliance matrix completed and accepted.
Procurement and Logistics Completion	04/05/2026	Buying Department / Constructor / Project Leader	Logistics and delivery plan completed, access permits confirmed, and safety method statements approved, allowing installation to start.
Installation and System Commissioned	19/05/2026	Constructor / Radio Engineering	Cabling, base stations, antennas, Wi-Fi APs, service configuration, and commissioning completed, with the system ready for validation tests.
Testing Validation Passed	25/05/2026	Radio Engineering	Radio, Wi-Fi, end-to-end connectivity, and port shutdown tests completed successfully, with issues closed or formally recorded.
Project Handover	01/06/2026	Project Leader / Radio Engineering	Final technical documentation delivered and formal handover/sign-off completed.
Project End	09/06/2026	Project Leader	Project administratively closed after the documentation closeout and planned risk-reserve period.

6 Organization and Communications

6.1 Organizational Breakdown Structure

The key stakeholders involved in the project are grouped into internal and external roles, as well as technical, business, construction and regulatory branches. This structure is summarized in the organizational breakdown structure shown in Figure 2, which will be used as a reference for the RACI matrix and cost model.

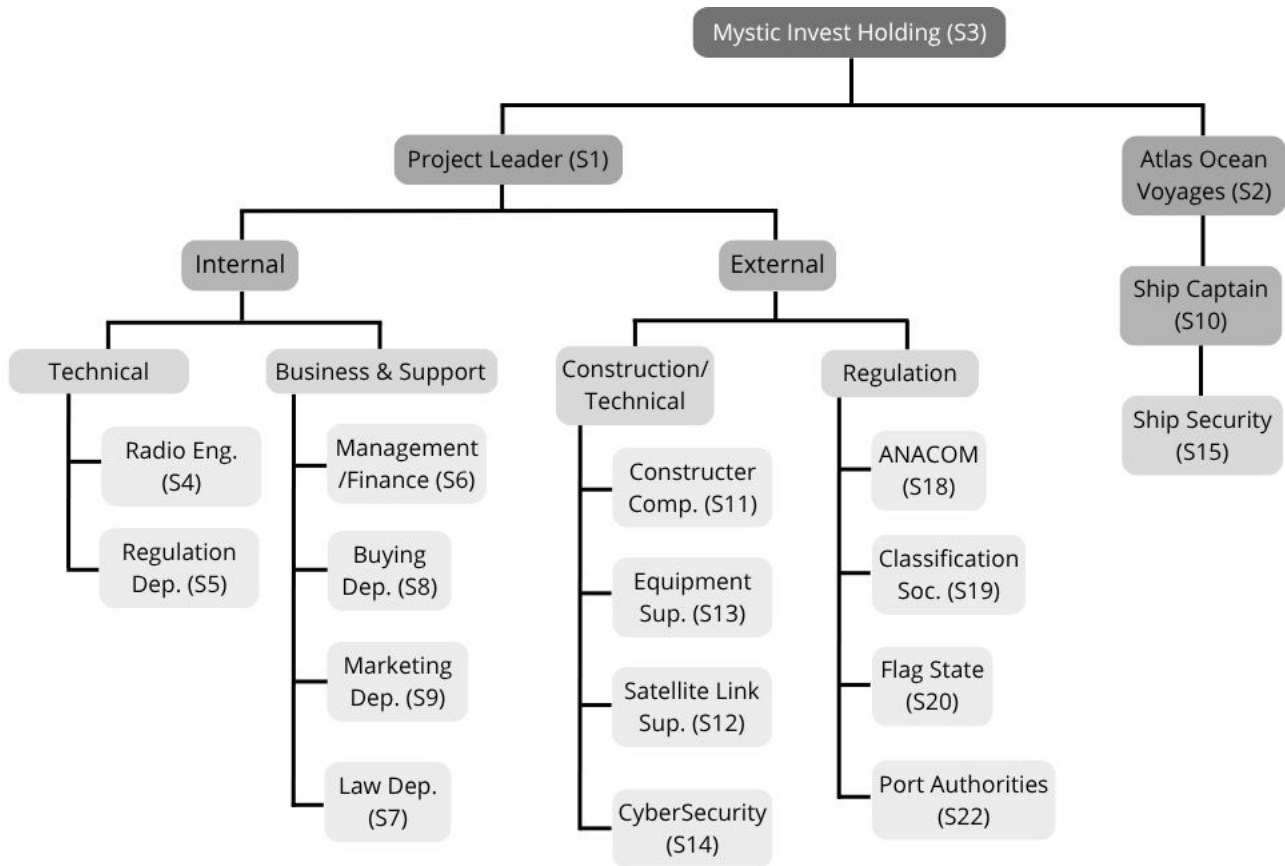


Figure 2: Organization Breakdown Structure for MV Voyager Telecommunications Project

6.2 RACI Matrix

The responsibilities of each stakeholder across the main project activities are defined using a RACI matrix. Table 5 specifies, for every WBS activity, who is Responsible, Accountable, Consulted and Informed, ensuring that ownership and decision paths are clear for the whole project.

Table 5: RACI Matrix

Code	Activity	Project Leader	Radio Eng. (Int)	Regulation Dep. (Int)	Business & Support (Int)	Constructor (Ext)	External Suppliers (Ext)	Client – Atlas Ocean	Ship Captain & Security	External Regulators (Ext)	Others
A	Gather Ship Data & Analyse Deck Layout	A	R			C		C	C (Security)	I	
B	Visit & Rank Candidate Locations	A	R			C		C	C (Capt. & Sec.)	I	
C	Coverage & Base Station Layout	A	R	C		C			I (Capt. & Sec.)		
D	Antenna & Cabling Paths	A	R	C		C		C	C (Captain)	I (Class. Soc.)	
E	Wi-Fi AP Layout & Service Zone Design	A	R	C	C (Marketing)	C	C (Cybersec.)	C			
F	Backhaul & Core Inter-connection	A	R	C			C (Cybersec. & Sat. Link)			I (Reg. Agencies) I (Class. Soc.)	
G	Cabling, Power & Telecom Room Design	A	R			C		C			
H	Hardware Technical Specifications	A	R	C			C (Equipment)			I	
I	Prepare Bill of Materials	A	R		C (Buying & Financial) C (Law)		C (Equipment)				
J	Set Roaming Parameters	A	C	R							I (Other Mob. Op.)
K	Design Port Shutdown Logic	A	R	C					C (Captain)	C (Port Auth.)	I (Other Mob. Op.)
L	Regulatory Conformity Analysis	A	C	R	C (Law)					I (ANACOM, Flag St.)	
M	External Approvals, Agreements & Compliance Matrix	A		R	R (Law)					C (all)	I (Other Mob. Op.)
N	Detailed Construction Package	A	R		C (Management)	C		I			
O	Access Permits, Planning & Safety Method Statements	A				R		C	R (Security)		
P	Logistics & Delivery Plan	A			R (Buying)	I	C (Equipment)	C	C (Capt. & Sec.)	C (Port Auth.)	
Q	Install Cabling, Power, Base Stations & Antennas	A	C			R		I	I (Capt. & Sec.)		
R	Install Wi-Fi APs & Network Equipment	A	C			R		I	I (Capt. & Sec.)		
S	Service Configuration & Shutdown Integration	A	R	C			C (Sat. Link, Cybersec.)				
T	System Commissioning	A	R			C		I	I (Captain)		
U	Radio, Wi-Fi & Load Testing	A	R		C (Marketing)			I	I (Capt. & Sec.)		C (Crew)
V	End-to-End Connectivity Test	A	R					I			
W	Port Shutdown Validation	A	R					I	C (Captain)	C (Port Auth.)	I (Other Mob. Op.)
X	Final Technical Documents Conception	A	R	C		C		C		I	
Y	Handover & Sign-off	R&A	R		I (Law & Fin.)			C			C (Insurance Underwriter)

Note: Columns are OBS groups; specific sub-stakeholders are named in parentheses. Roles merged from the original RACI; where two tasks were combined, the strongest role per group was kept.

6.3 Communication Plan

The communication plan ensures that the right information reaches the appropriate stakeholders at the right time and in the correct format. To manage the complexity of the MV World Voyager telecommunications deployment, communications are grouped into key categories: Executive Reporting, Internal Synchronization, Financial & Legal, On-Site Operations, Regulatory Compliance, and Market Intelligence.

Table 6 outlines the target audience, communication format, frequency, and the responsible owner for each communication stream.

Communication Stream	Format / Deliverable	Target Audience	Frequency	Owner
Executive & Sponsor Updates	Monthly executive summaries, milestone reports, and budget vs. actual tracking	Atlas Ocean Voyages (S2), Mystic Invest Holding (S3)	Monthly	Project Leader
Internal Team Synchronization	Weekly status meetings, progress reports, technical compliance docs, and UI/UX feedback	Project Dev. Dept. (S1), Radio Engineering (S4), Regulation Dept. (S5), Marketing Dept. (S9)	Weekly / As needed	Project Leader
Financial, Legal, Procurement & Risk	Bi-weekly financial reports, labor hour tracking, legal reviews, and risk assessment reports	Management (S6), Law (S7), Buying (S8), Insurance Underwriter (S21)	Bi-weekly / As needed	Management Dept. / Project Leader
Technical & IT Security	Technical data sheets, SLAs, bandwidth dashboards, and security audit reports	Sat. Link Supplier (S12), Equip. Suppliers (S13), Cybersecurity (S14)	Milestone-based / Continuous	Radio Engineering / Project Leader
On-Site Operations & Installation	Daily briefings, “Toolbox” safety meetings, direct radio contact, and access logs	Ship Captain (S10), Constructor (S11), Ship Security (S15)	Daily (during execution)	Project Leader / Constructor
Regulatory & Compliance	Formal license applications, EMI/EMC studies, certification submissions, and port entry notifications	Regulators (S18), Classification Society (S19), Flag State (S20), Port Authorities (S22)	Milestone-based	Regulation Dept.
Commercial, Roaming & Market Intelligence	Roaming contract negotiations, geofencing coordination, and monitoring competitor filings	Roaming Partners (S23), Competitors (S24)	Continuous / As needed	Marketing Dept. / Management
End-User Communications	Welcome brochures, digital “How-To” guides, and internal memos via ship’s bulletin	Passengers (S17), Crew (S16)	Project Closeout	Marketing Dept.

Table 6: Communication Plan

7 Resources and Cost Management

The project resources were grouped into labor, consultancy, equipment, software, and logistical inputs to support both the resource plan and the cost estimate. Table 7 summarizes the available resources and their standard rates in the MS Project resource pool.

Table 7: Project resource pool and standard rates.

Resource Name	Type	Max. Units	Std. Rate
Project Leader	Work	100%	120,00 €/hr
Radio Engineer (Sr)	Work	100%	120,00 €/hr
Law Officer	Work	100%	120,00 €/hr
Radio Engineers	Work	200%	100,00 €/hr
Regulation Officers	Work	100%	100,00 €/hr
Finance / Management Analyst	Work	100%	100,00 €/hr
Buying Officer	Work	100%	80,00 €/hr
Marketing Specialist	Work	100%	80,00 €/hr
Constructor Fee	Work	500%	100,00 €/hr
Cybersecurity Consulting Fee	Work	100%	120,00 €/hr
Regulatory & Licence Fees	Material	—	3.500,00 €
Classification Society Certificates	Material	—	2.300,00 €
Shipping & Logistics	Material	—	1.000,00 €
Satellite Sign-up Fee	Material	—	5.000,00 €
Indoor Cellular Base Station	Material	—	3.000,00 €/pu
Maritime Wi-Fi 6 Access Point	Material	—	50,00 €/pu
Indoor Antenna	Material	—	80,00 €/pu
Deck / Outdoor Antenna	Material	—	30.000,00 €/pu
Structural Cable (CAT6A / Fibre)	Material	—	5.000,00 €/pu
Network Switch / Gateway	Material	—	50,00 €/pu
Power & UPS Module	Material	—	2.500,00 €/pu
RF Propagation Software	Material	—	5.000,00 €

Table 8 then maps those resources to the project activities, showing the associated cost, duration, and allocation profile used for the effort and cost analysis.

Table 8: Activity cost and resource allocation matrix. **Total work package cost estimate is 214 476,00 €**

Activities	Cost (€)	Dur.	Project Leader	Radio Eng. (Sr)	Radio Engineers	Constr. Fee	Regulat. Officers	Law Officer	Cybersec. Consult. Fee	Market. Specialist	Finance / Mgmt Analyst	Buying Officer
PM	13 560	56.5 d	25%									
A	4 368	3.5 d		30%	100%	20%						
B	1 760	1 d		100%		100%						
E	2 672	2 d		10%	100%	5%	10%		20%	20%		
C	5 664	4 d		10%	150%	5%	10%					
D	2 832	2 d		10%	100%	30%	10%					
F	4 368	3 d		10%	100%		10%		50%			
G	7 080	5 d		10%	150%	15%						
H	2 760	3 d			100%		15%					
I	2 016	2 d		30%							50%	50%
J	2 176	2 d		15%			100%	15%				
K	5 976	3 d		70%	150%		15%					
L	2 176	2 d		10%			100%	20%				
M	11 080	3 d					100%	100%				
N	2 080	2 d			100%	10%					20%	
P	2 280	5 d										40%
O	1 600	2 d				100%						
Q	96 908	7 d		15%		300%						
R	12 976	4 d		15%		200%						
S	9 032	3 d		20%	100%		20%		20%			
T	1 040	1 d		100%		10%						
U	6 632	1 d		20%	100%					100%		
V	992	1 d		20%	100%							
W	1 984	2 d		20%	100%							
X	9 024	4 d		10%	200%	10%	60%					
Y	960	1 d		100%								

8 Schedule and MS Project Integration

8.1 Pert Diagram

The project is highly sequential, with most activities lying on the critical path (Figure 3):

M1 → A → B → C → F → G → M2 → H → I → L → N → M4 → P → O → M5 → Q → S → T → M7 → U → V → W → M8 → X → Y → M9

However, minor scheduling flexibility exists within the design, compliance, and installation phases. Specifically, Task E offers the highest buffer with 5 days of total (and free) slack, followed by Task M and Milestone M3 with 4 days. Tasks J and R provide 3 days of slack, Task K provides 2 days, and Task D has a minimal buffer of 1 day.

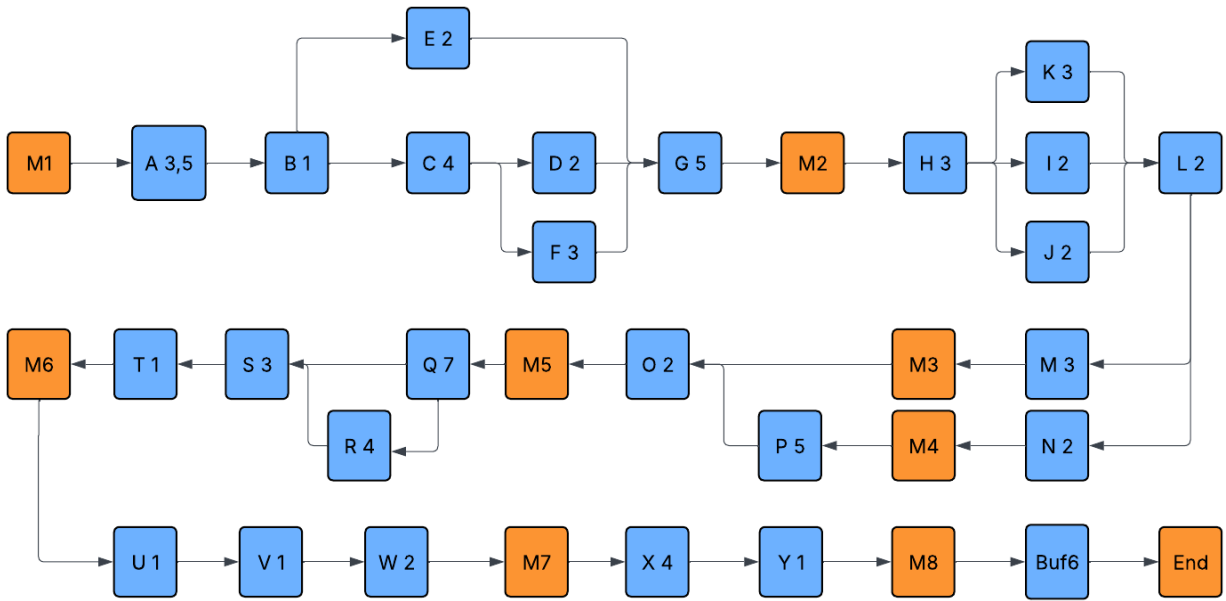


Figure 3: PERT Chart: Network Diagram in AON format

8.2 Gantt Chart and Resource Leveling

Given the technicality of the project, the most sought-after resources since the early beginning, were the Radio Engineers. Over-allocation of these resources lead to the application of resource leveling techniques. The Gantt chart (Figure 4), reflects the final schedule, with resource leveling. No overtime was used, instead, the following adjustments were made:

- **Task E** was delayed 2 days within its available float;
- **Task C** had its resource loading unbalanced so as to take advantage of the delaying of task E.
- **Tasks H and K** also required resource unbalancing, as both depend heavily on the same resource.

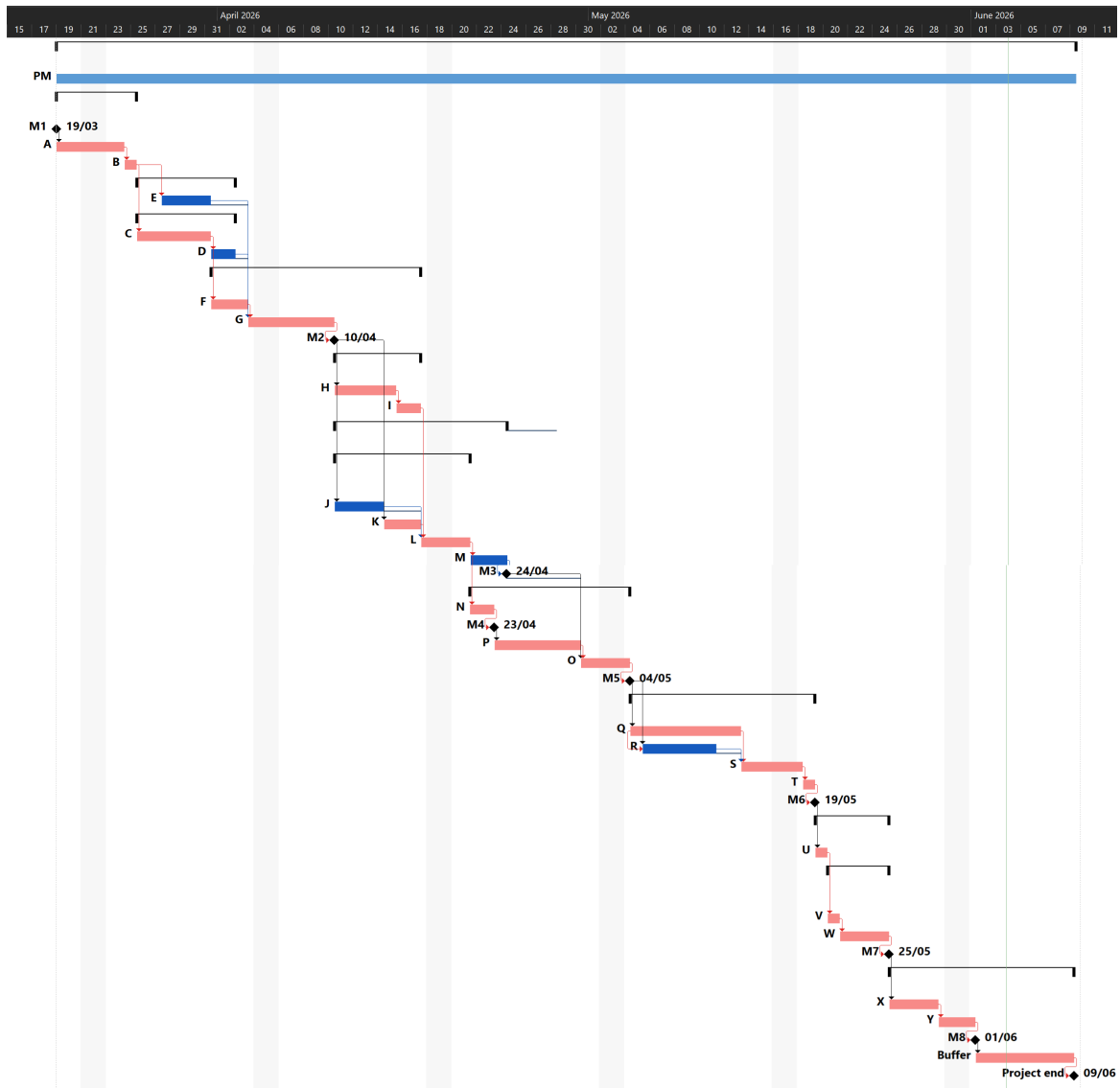


Figure 4: Gantt Chart: Project timeline and task dependencies.

- **Task R** was moved later in the schedule (Start-to-Start lag of 1) and had its duration increased from 3 to 4 days to free Radio Engineer capacity for the critical Task Q
- **Task K** was initially non-critical. By introducing a Finish-to-Start lag of 2 days on the M2 → K dependency, the resource conflict was resolved; however, this lag consumed K's float entirely, making it a critical task.

Thus, now there are two critical paths, and so the risk of the project increased. To mitigate natural fluctuations and delays in project duration, a 6 working day buffer was introduced after task Y and before the project completion date. The total project estimated duration is thus **58,5 working days**.

9 Quality Control Plan

The quality control plan defines the acceptance thresholds used to verify that the proposed telecommunications solution for the MV *World Voyager* is technically coherent and ready for implementation planning. The controls are linked to the project requirements and are applied at six checkpoints: design review, regulatory review, procurement review, installation readiness review, testing review, and final acceptance review.

The main quality focus areas are directly traced to the requirement set. Coverage is verified against TR-01, capacity against TR-03, satellite backhaul against TR-04, port shutdown behavior against TR-05, equipment and room suitability against TR-02 and TR-06, and overall project documentation against PM-01 to PM-04. This traceability ensures that every control item has a clear acceptance target and a clear owner.

Table 9 summarizes the main quality control items, their target criteria, verification method, responsible party, and corrective action logic.

Table 9: Quality control plan

Quality item	Target / acceptance criterion	Verification method	Responsible	Corrective action if not compliant
Coverage design	At least 95% of cabins, public areas, exterior decks, and operational zones covered by the planned cellular and Wi-Fi solution; no uncovered area larger than 5% of a deck surface (TR-01)	RF planning review and coverage maps	Radio Engineering	Revise AP/base-station placement, antenna orientation, or propagation assumptions
Capacity support	Planned capacity for 325 simultaneous users with at least 20% headroom in high-traffic areas; busy-hour load supported without overload (TR-03)	Capacity estimation and traffic review	Radio Engineering / Project Leader	Rebalance service zones, increase AP density, or revise user-loading assumptions
Backhaul integration	End-to-end design supports satellite backhaul with a target one-way latency below 700 ms and packet loss below 1% in the planning assumptions (TR-04)	Architecture review and connectivity validation	Radio Engineering / Satellite Supplier	Revise interconnection design, service policy, or link assumptions
Port shutdown behavior	Cellular transmission must be disabled in port within 30 s of shutdown activation and remain off until release is confirmed (TR-05)	Regulatory review and shutdown logic validation	Regulation Dept. / Law Dept.	Redesign shutdown logic and update compliance documentation
Equipment conformity	All selected network components must be marine-grade and compliant with the approved bill of materials; equipment-room occupancy below 85% of available rack space (TR-02, TR-06)	Datasheet inspection and procurement review	Buying Dept. / Radio Engineering	Replace non-compliant components, revise specifications, or reduce equipment footprint
Installation readiness	Installation package complete; drawings, access windows, permits, and safety method statements approved before execution (PM-04)	Installation readiness review	Project Leader / Constructor	Update drawings, routing, access planning, or safety documents
Final documentation	Final package delivered with 100% of mandatory artifacts (PM-01, PM-02, PM-04).	Final document review	Project Leader	Complete missing items and correct inconsistencies before handover

EVM is used in parallel to monitor whether the project remains within the planned cost and time boundaries. A bi-weekly review is sufficient for this project because the schedule is short and the key technical decisions are tightly coupled. At each review point, the team compares

planned value, earned value, and actual cost, then checks whether any quality deviation is also affecting the schedule or budget baseline.

10 Risk Management

Risks are assessed using the project’s standard matrix, evaluating the Probability of Occurrence and the Impact on Results (that takes into account cost and time increases as well as requirement fulfillment). The intersection of these metrics determines the final Type of Risk on a scale of 1 to 6. Table 11 presents the risk response plan.

Prob. of Occurrence		Impact on Results		Risk Matrix (Probability × Impact)																														
Probability				Very Low	Low	Moderate	High	Very High																										
Very High (5)	70%–95%	Very High (5)	Very High								2	3	4	5	6		1 – Negligible																	
High (4)	50%–70%	High (4)		High	2	2	3	4	5									2 – Minor																
Moderate (3)	30%–50%	Moderate (3)																	Moderate	2	2	3	3	5		3 – Moderate								
Low (2)	10%–30%	Low (2)																									Low	1	2	2	3	4		4 – Significant
Very Low (1)	5%–10%	Very Low (1)																																
										6 – Catastrophic																								

10.1 Contingency Reserve

The contingency reserve was sized bottom-up using an Expected Monetary Value (EMV), so that the reserve reflects the project’s actual risk profile. For each identified risk, the cost and schedule impact *if it occurs* were estimated from its contingency action and weighted by the probability of occurrence.

The reserve is the sum of these expected values, reported per risk in Table 10¹.

The cost-if-occurs figures are traceable to the project’s own cost model: labor is priced from the relevant installation task’s daily rate and hardware at the bill-of-materials unit prices. An inefficiency factor is applied to remedial labor to reflect the lower productivity of rework. As no spare hardware is held on board, a procurement lead time of two working days is added to the schedule impact of risks that require re-ordering. The reserve also reflects each risk’s chosen response strategy. The avoided risk (R-02) contributes nothing, as the threat is eliminated. The transferred risk (R-07) contributes no cost but retains a residual schedule impact, since insurance compensates for money and not for lost time. The accepted risks (R-08, R-09) are deliberately funded from the reserve: prevention was judged more expensive than the small expected cost of occurrence.

Only delays on the critical path consume the schedule reserve. The aggregate expected schedule impact of approximately five and a half working days is therefore contained within the six-day buffer placed before project completion.

The resulting contingency reserve of 15 174€ (about 7% of the work-package estimate) is added to the baseline to form the cost baseline. A separate management reserve of 10% (22

¹Totals computed from unrounded EMV values; columns may differ by ±1 € due to rounding.

917€) of the cost baseline is held outside the baseline to cover unidentified, unquantifiable risks, giving the total project budget reported in Table 12.

Table 10: Contingency reserve – expected cost and schedule impact per risk.

Risk	Prob.	Cost if fired (€)	Delay if fired (d)	EMV cost (€)	EMV delay (d)
R-01	0.4	16 611	4.8	6 644	1.9
R-02	0.2	0 (avoided)	0.0	0	0.0
R-03	0.2	5 992	2.5	1 198	0.5
R-04	0.4	5 960	2.0	2 384	0.8
R-05	0.4	9 408	2.5	3 763	1.0
R-06	0.4	1 960	2.0	784	0.8
R-07	0.2	0 (transferred)	1.0	0	0.2
R-08	0.2	1 000	1.0	200	0.2
R-09	0.4	500	0	200	0
Total				15 174	5.4

Table 11: Risk Response

Identification			Characterization						Plans			Execution			
Code	Description	WBS	Cause	Prob.	Imp.	Importance	Mitigation	Contingency	Monitoring Indicator	Start	End	Responsible			
R-01 (Mitigate)	Severe signal attenuation.	C, E, U	Metallic bulkheads block RF propagation more than anticipated.	3	5	5 – Severe	Conduct 3D RF propagation simulations using computational models before locking the coverage design.	Re-run the 3D RF analysis, procure and install additional cellular hardware and additional Wi-Fi APs, then re-commission and re-test the affected zones.	Number of signal "dead zones" detected.	C	U	Radio Engineering			
R-02 (Avoid)	Port shutdown non-compliance.	K, L, S, W	The automated shutdown fails port regulatory standards, risking fines.	2	5	4 – Significant	Strategy: Avoid – Co-design the shutdown logic directly with regulators for pre-approval.	N/A (Threat eliminated in planning)	Port shutdown test results.	K	W	Regulation Dept.			
R-03 (Mitigate)	Satellite back-haul test failure	F, V	The selected satellite back-haul underperforms during project commissioning, failing to meet required latency or packet loss thresholds for handover.	2	4	3 – Moderate	Select a satellite provider with multi-orbit capabilities to ensure tracking redundancy.	Dispatch emergency vendor support for antenna realignment and lease a temporary high-capacity test frequency to achieve acceptance parameters.	Ping latency and packet loss %.	F	V	Sat. Link Supplier			
R-04 (Mitigate)	Hardware delivery delays.	P, Q, R	Supply-chain bottlenecks or extended lead times on long-lead maritime-grade hardware delay the start of installation activities.	3	3	3 – Moderate	Place hardware orders with an explicit 4-week buffer built into the delivery schedule.	Resequence the installation to use available work fronts while awaiting the late item. If re-sequencing cannot recover the schedule, expedite/air-freight the critical item.	Supplier delivery tracking status.	P	R	Buying Dept.			
R-05 (Mitigate)	Power quality/continuity risk to telecom racks	G, Q	The ship's onboard supply suffers voltage sags, transients or brief interruptions, resetting or damaging telecom equipment.	3	4	3 – Moderate	During the site survey, measure the onboard power quality at the planned telecom feeds and size the racks' UPS.	Procure and install standalone maritime UPS systems for the telecom racks.	Measured power quality vs. input tolerance	G	Q	Radio Engineering			
R-06 (Mitigate)	Regulatory approval delays.	L, M	Slow bureaucratic processing from Flag State or Classification Society.	3	4	3 – Moderate	Submit compliance matrices immediately after preliminary design finishes.	Reschedule the physical installation phase to later in the docking timeline.	Days pending for regulatory approval.	L	M	Law & Regulation Dept.			
R-07 (Transfer)	Damage to luxury interiors.	Q, R	Accidental damage by the constructor during drilling/cabling in premium passenger areas.	2	3	2 – Minor	Shift financial impact to the Constructor's insurance to cover luxury panel repairs.								
R-08 (Accept)	Cabling paths blocked.	D, G, Q	Existing cable trays are full, requiring new structural drilling.	2	2	2 – Minor	We understand the risk and choose to do nothing upfront.								
R-09 (Accept)	Minor aesthetic mis-alignment.	R	An installer mounts Wi-Fi APs slightly off-centre from the ceiling architect's grid, requiring a quick remount.	3	1	2 – Minor	We understand the risk and choose to do nothing upfront.								

11 Conclusions & Project Highlights

The project delivered a telecom deployment plan for the MV World Voyager within a duration of 58,5 days, covering the technical design, work breakdown structure, schedule, resource plan, risk analysis, and quality control framework. The final budget structure is summarized in Table 12, which shows the relationship between the work-package estimate, contingency reserve, cost baseline, management reserve, and total budget.

Table 12: Project Budget Breakdown

Budget Component	Amount (€)
Work-package cost estimate	214 476
Contingency Reserve (7%)	15 174
Cost Baseline	229 650
Management Reserve (10%)	22 965
Total Budget	252 615

Several decisions increased the technical rigour of the project and improved traceability across the planning package. The WBS was structured so that each deliverable maps directly to a clear group of activities, which simplifies the link between scope definition and execution planning. The schedule was refined through PERT analysis and resource leveling, which helped identify the critical path and resolve resource conflicts without using overtime.

The quality control framework also adds strength to the project because each acceptance criterion is linked to a specific requirement and corrective action. This makes the plan easier to validate and keeps the technical, regulatory, and documentation checks aligned with the original scope. In parallel, the risk analysis was developed using a probability-impact matrix and an EMV-based contingency reserve, giving the budget a more realistic treatment of uncertainty.

Another relevant aspect is the integration of stakeholder, communication, procurement, and compliance planning into a single coherent report. That integration is particularly important in a cruise-ship environment, where technical design is constrained not only by coverage and cost, but also by maritime safety, port regulations, and installation logistics. Overall, the project shows a realistic and structured approach to planning a telecom deployment in a complex marine setting.

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